Chapter 5

Aesthetic problems

David Bartlett

CHAPTER CONTENTS

Causes of aesthetic problems 55
The shape and size of teeth 55
Colour of teeth 56
Black triangles 59
Tooth position – spacing 59
Malformed teeth 61
Missing lateral incisors 62
The preoperative assessment 62
Bleaching 63
Vital bleaching 63
Non-vital bleaching 65
Whitening strips and topically applied systems 66
Acid microabrasion 66
Summary 67

The meaning of an aesthetic problem is vague as perfectly normal looking teeth can appear unacceptable to a patient. So the first stage in any assessment is to determine with the patient what the problem is and to assess what can be done. This chapter details the specific aesthetic problems a patient may present with and some simple solutions to address some of these, such as direct composite additions and vital and non-vital bleaching. Chapter 15 revisits some of the issues raised in this chapter and addresses how optimum aesthetics can be created in indirect restorations and describes in detail aspects such as tooth shaping and correct shade matching.

CAUSES OF AESTHETIC PROBLEMS

The shape and size of teeth

The teeth that are visible for most patients are the upper canines and incisors; however, in some patients, premolar teeth or even molar teeth are visible when the patient talks and smiles. The position of the lips in the extreme natural position of smiling is called 'the smile line' by many prosthodontists. The visible shape of teeth consists of the length, width and their relationship to other teeth. Generally, upper incisors are about 1 cm long and have a width between 0.75 and 0.80 cm (Figure 5.1). Although these dimensions vary considerably, teeth made too wide or too long appear unnatural (Figures 5.1 and 5.2). Therefore, as a base line, knowing the average tooth length and width can be useful in determining what is abnormal and what can be done to teeth in the smile line (see Chapter 15 for creating the ideal aesthetics and average tooth dimensions).

The other important criteria in an assessment of the smile line are the gingival contours, particularly along the margins of the upper anterior teeth. For some patients the smile line is low and the gingival margins are not visible; thus aesthetic anomalies are therefore relatively unimportant. In others, the smile line is high and gingival margins are clearly visible, variations from the norm having a high impact (Figure 5.3). In health, the gingival margins of the canines and central incisors are normally at the same horizontal level and more apical to the lateral incisors (Figure 5.4). The incisal edges of the upper central incisors and the tips of the canines are also usually at

© 2011 Elsevier Ltd.



Figure 5.1 Missing upper lateral incisors with a spaced anterior region. The central incisors and canine teeth are already approaching the limit of their width.



Figure 5.2 The addition of composites to the patient in Figure 5.1 has widened the teeth beyond an acceptable level, leading to less than ideal aesthetics.



Figure 5.3 The crowned upper right lateral incisor has fractured in the past and overerupted, bringing with it the alveolar tissues, leading to an unsightly appearance at the gingival margin.



Figure 5.4 The gingival margins of the lateral incisor teeth are placed at a slightly lower position than the centrals and the canines.

the same approximate level but the lateral incisors are normally slightly shorter. The horizontal line along the incisal edges from the incisors to the canines is normally convex and follows the line of the lower lip (Figure 5.5).

Whilst these guides can be helpful they are not prescriptive and individual variation occurs. However, they are worth remembering when trying to assess a patient's smile when problems are present.

Colour of teeth

A frequent patient complaint regarding dental aesthetics is that teeth tend to discolour or darken with increasing age. The reason for this change is believed to be caused by the gradual thinning of enamel from tooth wear. As the

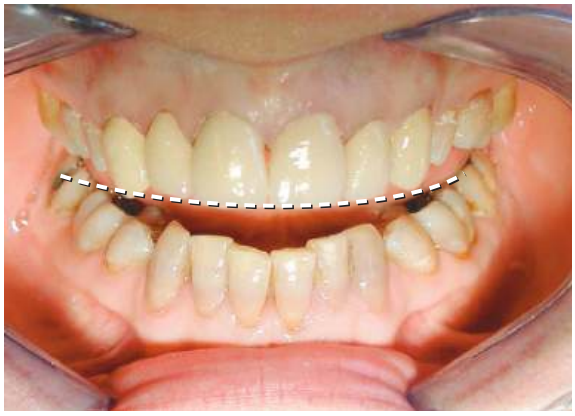


Figure 5.5 The incisal tips of the upper central incisor and canine are at a similar level on the patient's right-hand side but not on the patient's left, where they are more uneven. The ideal relationship is shown on the patient's right side. However, the gingival margins on the left side are present at the most appropriate level, whereas on the right they are not.



Figure 5.6 The canines are slightly darker than the other teeth. This natural difference between the canines and incisor teeth is sometimes unacceptable to patients and requires bleaching.



Figure 5.7 **A** Dental fluorosis in this patient has produced an unsightly white and brown mottled appearance. **B** Severe dental fluorosis in this patient has produced an unsightly brown pitted appearance.

enamel wears, the darker dentine shines through and becomes more dominant in the colour of the teeth. An unacceptable colour is personal and subjective. What might appear to be a perfectly normal colour may be unacceptable to a patient. A common example of this is the colour of the canine tooth (Figure 5.6). Naturally, this tooth has a higher saturation of the colour yellow than other incisor teeth and the difference in colour is unacceptable for some patients.

Assessing what is wrong with the colour is challenging and needs careful thought and considered planning time. Generally, if someone wants their teeth to be whiter the treatment options usually at first may appear to be relatively simple. But it is worthwhile taking time to ensure that the colour is the only reason for their dissatisfaction. It is not unusual for patients to focus on one aspect of their appearance while ignoring the others, only to find that after treatment of one problem, the other problems come to the fore. During the assessment of colour it is useful to enquire about previous photographs and use them as a guide to identify the problem.

Occasionally, more serious discolorations of teeth present. This may be a result of fluorosis (Figure 5.7) or tetracycline staining (Figure 5.8). The former can present in patients born overseas where there is a high fluoride content found naturally in spring water, or those given too high a dose of fluoride during tooth development. Our understanding of the action of fluoride is now clearer and the most important preventive effect is from its topical application, systemic fluoride supplements such as drops or tablets are rarely given today due to the disfiguring white and brown mottled defects that can occur (Figure 5.7).

Tetracycline staining is very unlikely to occur to someone born in the UK as the disfiguring side effect of tooth discolouration is well known (Figure 5.8), hence it is



Figure 5.8 Tetracycline staining.

rarely given to children during tooth development. For those born outside the UK the chances of developing tetracycline staining will depend on the availability of the drug over-the-counter. In some parts of Asia, for example, drugs which would be prescription delivered in the UK are

available over-the-counter and this is particularly true of antibiotics. Tetracycline staining can be clearly distinguished as the teeth fluorescence under ultraviolet light.

Genetic disorders such as dentinogenesis and amelogenesis imperfecta also affect tooth development, with both having varying impact upon the teeth. Mild forms of either condition can be difficult to distinguish but more severe forms are characteristic and differentiated by the anatomy of the roots. In dentinogenesis imperfecta there are bulbous roots with root canal obliteration which are visible on radiographs (Figure 5.9), whereas teeth with amelogenesis imperfecta have normal root formation but abnormal enamel development.

Dentinogenesis imperfect (Figure 5.9) is inherited as an autosomal dominant trait, with one parent usually having the condition. The disorder is the result of a defective DSPP (dentine sialophosphoprotein) gene. Three types have been identified: Type I is associated with osteogenesis imperfecta, whilst Type II (some have progressive hearing loss) and Type III are not associated with other inherited disorders and some believe they form a single disorder. It can affect both primary and permanent dentitions. The crowns of teeth affected by dentinogenesis imperfecta appear blue–grey or yellow–brown and translucent; the condition has also been referred to as hereditary opalescent dentine. The defective dentine leads to early loss of the enamel and rapid wear.

Numerous forms of amelogenesis imperfecta have been described based upon the appearance of the teeth and the way in which the disorder is inherited; most cases are inherited as an autosomal dominant trait. Amelogenesis imperfecta results from mutations in the genes that are responsible for the production of proteins that are important in the formation of enamel. It can also affect both primary and permanent dentitions. The teeth are usually discoloured with a pitted or grooved appearance and can



Figure 5.9 Clinical images of a patient with dentinogenesis imperfecta showing yellow–brown discolouration and translucent appearance (A) and the rapid wear in a different patient that can occur due to early loss of enamel (B). The radiographs (C) show bulbous roots with root canal obliteration.

(Courtesy of Scully C, Flint S, Porter SR, Moos K. Oral and maxillofacial diseases, 3rd ed. London: Taylor & Francis, 2004)



Figure 5.10 This patient has a form of amelogenesis imperfecta leading to rapid wear of the teeth (see inset).

break and wear down rapidly (Figure 5.10). Some forms of amelogenesis imperfecta can be difficult to distinguish from fluorosis (see Figure 5.7), as both conditions can lead to hypoplastic changes to the surface of the enamel.

Due to the often rapid wear of teeth with either dentinogenesis or amelogenesis imperfecta, early intervention with multiple crowns is often required to prevent the teeth from wearing to a degree that would prevent restoration without resorting to removable prosthodontics.

Black triangles

One of the outcomes of periodontal disease is gingival recession. As the gingival margin migrates apically past the widest aspect of the roots (at the crown–root junction), the interdental space takes on a triangular appearance due to the tapering of the roots of anterior and premolar teeth. The crest of the gingiva interdentally will therefore have a wide base and the space between the teeth will have a point towards the contact point – the so-called black triangle (Figure 5.11). A reduction in the extent and position of the interdental papilla is challenging and a suitable solution to address the situation may not be feasible or stable following treatment. Widening the teeth either side of the black triangle to reduce the gap may create teeth with a poor height to width ratio and an unnatural appearance. Another disadvantage of increasing the width of teeth is that the increase in the bulbosity of the tooth at the gingival margin changes the emergence profile and increases the difficulty in cleaning. Continued poor oral hygiene in someone with a history of periodontal disease may increase the risk of further recession and deterioration in any improvements made to the 'black triangle'. Therefore, if changes to the width of teeth are considered it is worthwhile preparing a diagnostic wax-up to assess the planned changes and the impact this will have on the appearance and health of the periodontal tissues.



Figure 5.11 The development of some localized periodontal disease and inappropriately shaped upper left central incisor crown has left a black triangle between the two central incisors. This patient also has tetracycline staining.

Tooth position – spacing

Diastemas and generalized spacing between teeth result from a mismatch in tooth and jaw size, which can be unacceptable to patients (Figure 5.12). Closure of diastemas and spaces can be accomplished by orthodontic movement or by restorations. Any orthodontic movement is unlikely to be stable over long periods of time and some degree of retention, using either a removable appliance for nighttime wear or a fixed wire retainer, is inevitable.

Restorative intervention can be a useful compromise but the crucial assessment is the width of the teeth. Naturally narrow teeth will accept an increase in width (Figures 5.12 and 5.13) but broader ones may appear to be too wide after any change (see Figure 5.2). A diagnostic wax-up will assist in the preoperative assessment (Figure 5.14) but a composite 'mock-up' may also be useful. Composites can be added to teeth without acid etching and bonding, shaping to a diagnostic wax-up and stent or building up



Figure 5.12 Unsightly dental spacing has resulted from relatively small teeth on large dental bases.



Figure 5.13 The spaces for the patient seen in Figure 5.12 have been successfully closed using composite restorations and no orthodontic treatment was necessary.



Figure 5.14 Diagnostic wax-up for the patient seen in Figures 5.12 and 5.13.

freehand. This can provide the patient with a realistic opportunity to see the proposed changes to shape and the colour match possible (Figure 5.15). If this is done without a diagnostic wax-up, and the patient is happy with the appearance achieved, an alginate impression of the result should be taken so that it can be copied in the definitive restorations which may be carried out some time later. These composite 'mock-ups' are easily removed once the patient has had sufficient time to assess the changes made by simply using an excavator at the margins of the composite to lever them off the tooth. Care must be taken to avoid the patient swallowing or inhaling the veneers of composite by using high volume suction lingual to the site and using a gauze or Spontex sponge to protect the oropharynx.

When making significant changes with composite it is useful to use a putty stent of a diagnostic wax-up or of a model made from an impression of the composite mock-up. Figure 5.16 shows how this was done for the patient in Figure 5.12.



Figure 5.15 This patient was unhappy with the alignment of the upper anterior teeth (A). A composite 'mock-up' (B) without etching the tooth and using bond allows the patient to assess the change in shape and colour possible prior to making a decision as to whether to proceed with the definitive composite additions.

Orthodontic treatment is indicated for more severe changes to the normal position of teeth. The patient must fully appreciate that orthodontics is the ideal treatment for moving teeth and that restorative intervention is often a significant compromise and generally involves at least composite build-ups, veneers or crowns. The irreversible nature of restorative treatment means that the patient must be fully committed and understand the risks involved with the proposals.



Figure 5.16A Addition cured polyvinyl siloxane putty impression taken of the palatal surface of the diagnostic wax-up seen in Figure 5.14. The putty stent is cut with a scalpel along the incisal edges of the teeth. This is then inserted into the patient's mouth and acts as a guide for the composite additions.

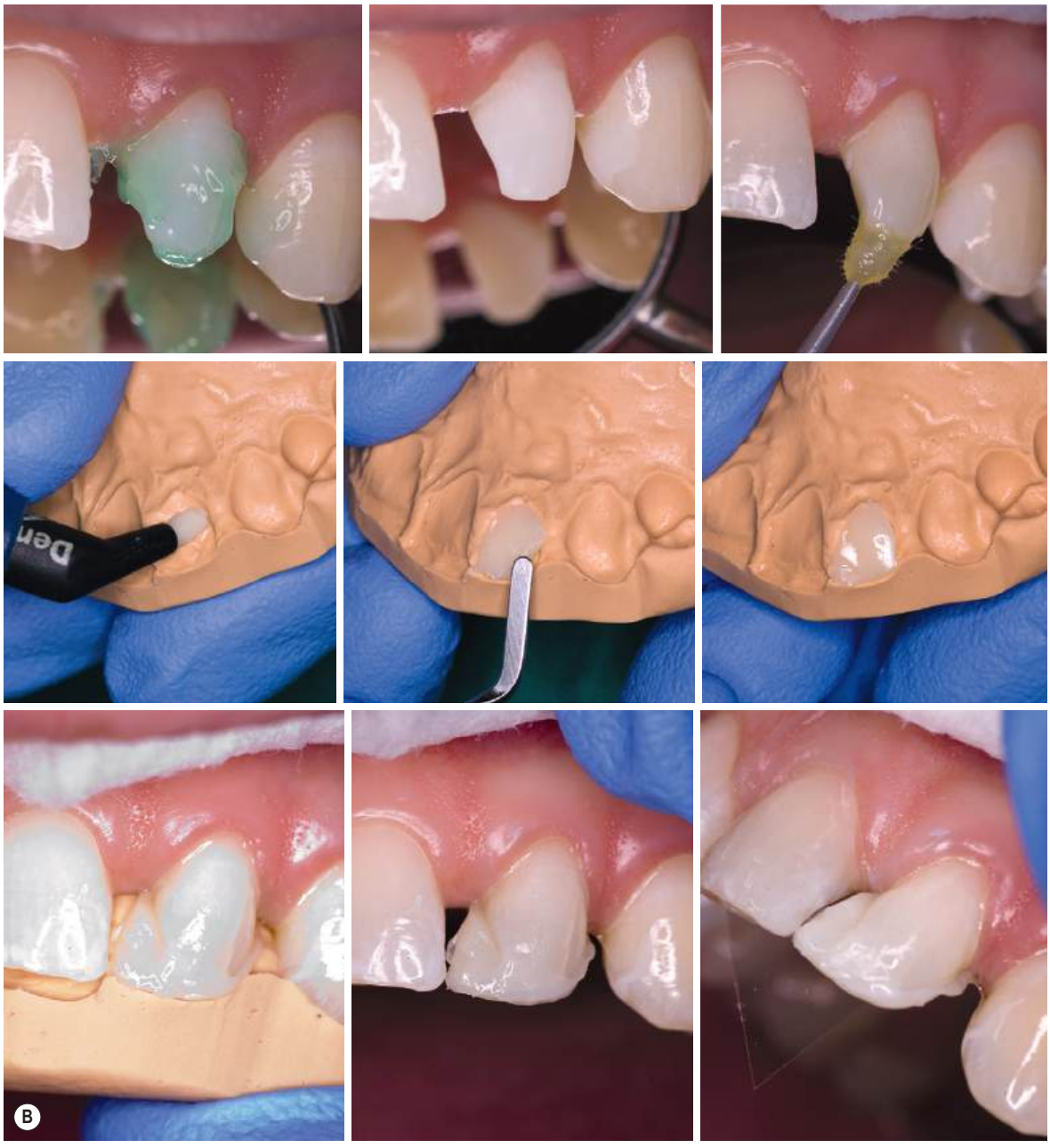


Figure 5.16B From top left to right: peg lateral all surfaces acid etched, rinsed and dried to give frosty appearance and bond applied. Composite added to stent and adapted with a flat plastic instrument. Stent reseated intraorally, ensuring composite adapted palatally and not contacting the adjacent teeth, the composite is now light cured – with stent removed cellulose strip inserted to allow proximal surface to be built up – labial face can now be completed (Figure 5.13).

Malformed teeth

It is relatively uncommon to find malformed teeth, particularly in the upper anterior region. Trauma to the deciduous teeth may result in a malformed permanent successor crown which only becomes visible once the tooth has erupted. Peg laterals are more common and may present with other alignment problems which may form part of an overall orthodontic treatment. Restorative management can be with either composite or porcelain veneers or occasionally crowns.

Missing lateral incisors

Hypodontia is relatively common, particularly missing third molar teeth, but the most commonly affected tooth which affects the appearance is the lateral incisor. This is normally recognized early in the adolescent's development and orthodontic treatment is used either to close the space, masking the canine shape with composite, or to create sufficient space for a replacement lateral incisor using either a minimal preparation bridge or an implant. More severe forms of hypodontia, with multiple missing teeth, create more restorative challenges which are outwith the scope of this book.

THE PREOPERATIVE ASSESSMENT

The first stage in addressing a patient's concern over appearance is to assess whether restorative treatment only is the ideal option. In many instances, particularly those involving tooth positions, orthodontic management may be the preferred option with or without subsequent restorative treatment. When restorative treatment is planned, routine caries and periodontal treatment and other disease control needs to be completed and followed by a reasonable period of stabilization to ensure that there is no new disease (these issues have been addressed in the preceding chapters). This period is not fixed and will to some extent depend on the severity of the problem, but at least 6 months without new disease and evidence of good oral hygiene and compliance may be sufficient.

Following disease control the next stage is to assess which restorative option is acceptable to the patient. A diagnostic wax-up is an invaluable tool to show patients the planned treatment and might be linked to a composite 'mock-up'. If a combined orthodontic restorative approach is considered, a Kessling set-up can be carried out, where teeth are sectioned from a model and moved into the anticipated position that orthodontic treatment could achieve and any restorative work being produced in wax or acrylic teeth (Figure 5.17). In most circumstances the diagnostic wax-up is simple and needs a single set of models; however, in more complex cases where the end result is not easily envisaged, more than one diagnostic wax-up is made to show the patient various options (Figure 5.18). This may also be linked to provisional restorations, preferably before the definitive treatment is provided.

When the only complaint is colour in unrestored teeth, the treatment can be a straightforward case of providing bleaching. For more extensively restored teeth or very discoloured teeth, veneers (see Chapter 12) or crowns are

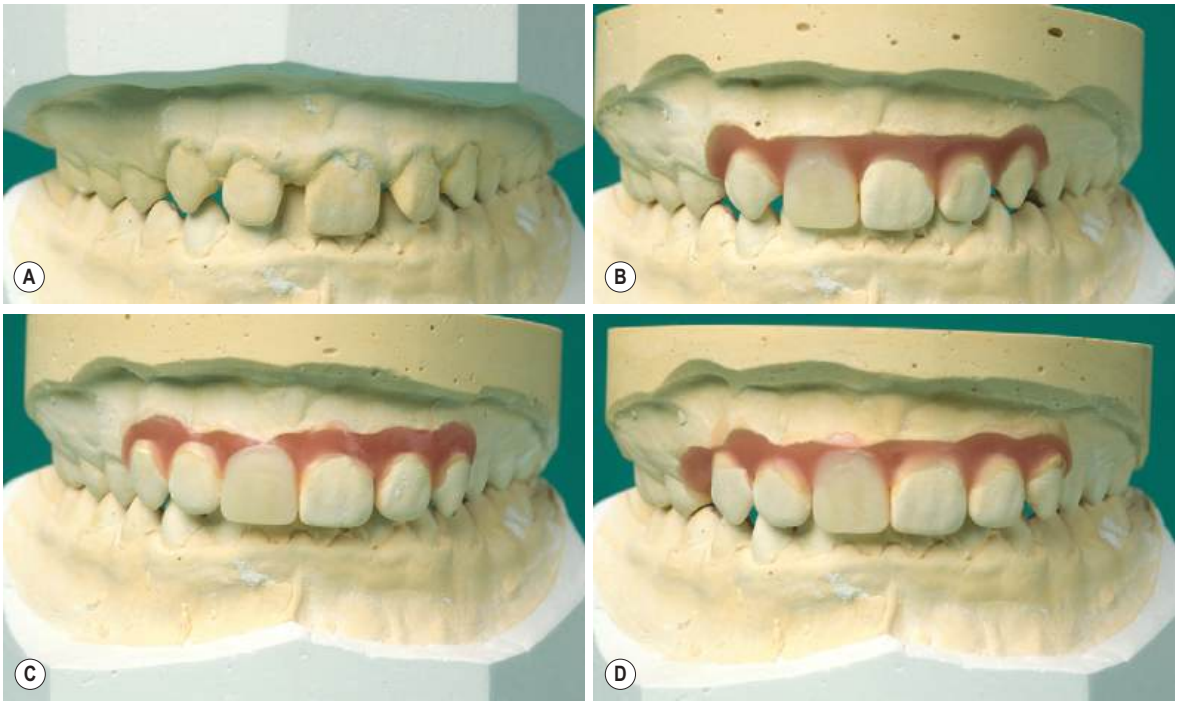


Figure 5.17 Kessling set-ups for a patient with a missing upper right central incisor (A). The treatment options explored are: (1) no orthodontics and build up the upper right lateral incisor with composite (B); (2) create space orthodontically for the upper right central incisor replacement (implant or minimum preparation (resin retained) bridge) without extractions (C) and with upper right premolar extraction (D).

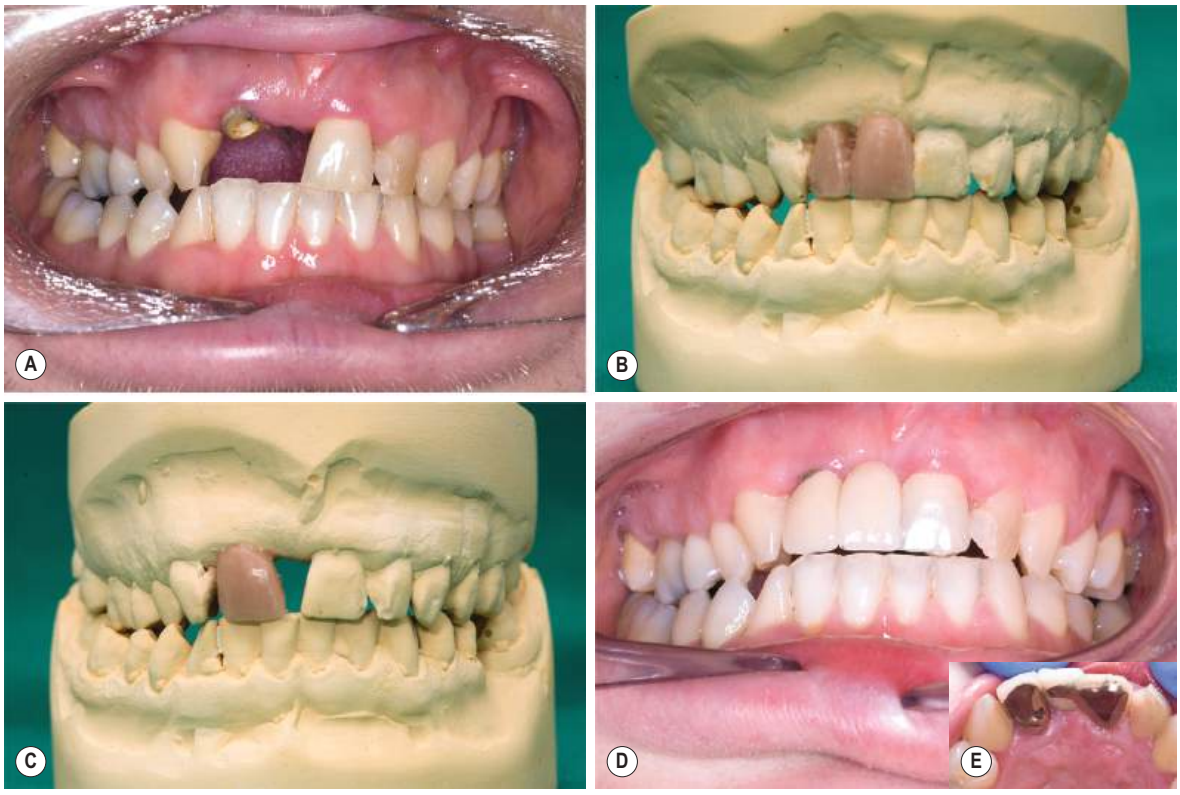


Figure 5.18 This patient has a missing upper right central incisor and retained upper right lateral incisor root. The diagnostic wax-ups show the two possible treatment options and allow the patient to decide on the favoured outcome. A cantilever minimum preparation, resin retained bridge off the upper left central incisor and a post crown on the upper right lateral incisor were provided.

indicated (see Chapters 10 and 11 on metal–ceramic crowns and all-ceramic crowns). Note that if veneers are placed around existing extensive restorations the risk of failure around the margins increases as an inferior bond may result. For those teeth with a problem with shape but not tooth position, a veneer made in either direct composite or porcelain can be the solution. In cases when changes to the tooth position and tooth shape are needed, minimal intervention techniques are often inappropriate and crowns are required. Before embarking upon more extensive restorations, particularly those involved with tooth position, an orthodontic approach must be considered first.

BLEACHING

Vital bleaching

Hydrogen peroxide (H_2O_2) has been used to whiten teeth for over a century. The molecule is minute and able to pass through the enamel to affect the dentine. It is believed that most of the action of bleaching is on the dentine rather than the enamel which is much more translucent. The mode of action is not fully understood but the small size of the molecule penetrates the dentine and removes stains from the tissue. The high reactivity of hydrogen peroxide and the subsequent release of the superoxide ion are understood to be the reason why it is so effective but how the stain is removed is less clear.

The most commonly available form of hydrogen peroxide is carbamide peroxide, otherwise known as urea peroxide, which breaks down to form hydrogen peroxide and urea. The hydrogen peroxide forms the superoxide ion and water whilst the urea breaks down to form ammonia and carbon dioxide. The urea formed as part of the breakdown is too small to have any toxicological consequence. There have been no clinical studies suggesting bleaching has any serious systemic risk other than that normally encountered with restorative materials. The effects on restorative materials themselves is largely transient and reversed within 24 hours but most practitioners recommend a few days between completion of bleaching and any restorative work involving bonding.

The main advantage of bleaching is that it is conservative of tooth tissue. The bleaching product can be delivered at the chairside or at home. Both techniques utilize

Table 5.1 List of stains successfully managed with bleaching. All require either unrestored or minimally restored teeth. Some of the acquired forms may be simply removed by scaling and polishing the teeth

ACQUIRED	DEVELOPMENTAL (MILD FORMS)
• Mild generalized staining • Age-related yellow discoloration • Tobacco staining • Dietary stains, e.g. tea and coffee • Colour change related to pulpal trauma	• Fluorosis • Tetracycline staining • Amelogenesis imperfecta

the same chemistry, namely the use of hydrogen peroxide. Chairside bleaching is particularly suitable for patients without the motivation for home care and who prefer dentists to manage their treatment. It is generally more expensive in surgery time and often more costly to patients but the result from both techniques is likely to be the same. Table 5.1 shows those cases most successfully treated with vital bleaching.

For home care a vacuum-formed splint is made to overlay the teeth. It is generally cut back just below the gingival margins and usually has no palatal coverage. It is loaded with bleaching agent by the patient and worn each night for 2–3 hours (Figure 5.19). In most superficial discolorations the results are seen within a couple of weeks (Figure 5.20). For more intense stains, particularly those associated with tetracycline or fluorosis, the bleaching may need considerably more time. It is worth using a shade guide to determine the colour before treatment and to monitor if and how well the bleaching is working.

The gingival margin area around teeth can be more resistant to bleaching and it is worthwhile warning patients if a particularly darkened area along the cervical margin exists (Figure 5.20C). The colour stability of bleached teeth is also not predictable as people respond differently to bleaching. Generally, teeth that have bleached well can remain stable for 2–3 years, after which further bleaching may be required. Patients should be warned about this potential risk prior to commencing treatment.



Figure 5.19 The bleaching tray is made over the teeth with a vacuum-formed plastic. Some technicians paint a spacer over the teeth to be bleached but this is not essential.



Figure 5.20A Two changes are needed to improve the appearance. Firstly to orthodontically move the teeth to remove the overlap and then bleaching to improve the colour.



Figure 5.20B Patient seen in Figure 5.20A. Following orthodontics the teeth are better aligned.



Figure 5.20C Patient seen in Figure 5.20A and 5.20B. Following bleaching the teeth have whitened but the gingival margin on the upper left central incisor has been more resistant.



Figure 5.21 An extensively restored and discoloured tooth unsuitable for bleaching. A new restoration (core) and crown would be more appropriate.

The most common side effect of vital bleaching is dentine sensitivity as the bleach has a low pH and opens up dentinal tubules. Generally, the symptoms are transient and do not threaten the pulp. Intermittent use or using desensitizing toothpastes should provide relief from pain for most sufferers, allowing the bleaching to continue, but in some cases the intensity of the pain prevents further bleaching.

The choice between bleaching and other options depends on the restorative status of the teeth and the severity of the discoloration. Extensively restored teeth or severely discoloured teeth (Figure 5.21) are more appropriately treated with crowns, whereas bleaching can be tried in teeth with less severe defects and restorative status.

Non-vital bleaching

Non-vital bleaching following successful endodontic treatment may avoid the need to use a crown to improve the appearance of anterior teeth that are minimally restored (Figure 5.22). Bleaching virtually unrestored root-filled anterior teeth preserves the tooth tissue. However, cases involving extensive restorations (more than a small Class III or IV) and dark discolouration (Figure 5.23) may need crowns and often a post to retain the core (see Chapter 7). Bleaching will generally improve the brown discolouration which occurs once the vitality of teeth is lost. The darker the stain, the more resistant to bleaching the tooth becomes.

Before embarking on non-vital bleaching a periapical radiograph is needed to ensure a good and effective root canal treatment is present. The gutta-percha should be removed from the pulp chamber and root canal to below the gingival margin and sealed with a glass ionomer cement. These two factors ensure that no bleaching products enter into the periradicular tissues. Rubber dam should be placed to ensure protection of the soft tissues. Sodium perborate is commonly used in non-vital bleaching. It is another product of the hydrogen peroxide family which breaks down to form sodium metaborate, hydrogen peroxide and nascent oxygen. This material has been used for over half a century to bleach non-vital teeth using the 'walking technique'.



Figure 5.22A Palatal view of a less severely discoloured and fractured upper right central incisor which is more appropriately treated with a bleaching agent and a conservative composite restoration.



Figure 5.22B Labial view of less severely discoloured and fractured upper right central incisor seen in Figure 5.22A.



Figure 5.23 Severely discoloured and fractured upper right central incisor requires a crown.

Sodium perborate mixed with hydrogen peroxide is placed in the coronal cavity and packed against the buccal wall and then sealed in with a temporary restorative material for about 2 weeks (Figure 5.24). This may need to be repeated a number of times to achieve the correct shade. Some practitioners acid etch the inside of the pulp chamber to open up the dentinal tubules to ensure better penetration of the active bleach products; however, there is no evidence to support its efficacy. As an alternative to the walking bleach technique, following placement of a well-sealed restoration in the coronal aspect of the root canal, carbamide peroxide delivered in a splint can be used by the patient in a similar way to vital bleaching.

Where non-vital bleaching is concerned some have reported external cervical root resorption, but this is more likely to be a result of the trauma that led to the discolouration in the first instance. The risk of resorption as a result of non-vital bleaching can be reduced with an appropriate technique described above.



Figure 5.24A Gutta-percha has to be completely removed from the pulp chamber for non-vital bleaching. The gutta-percha should be covered with a layer of glass ionomer to provide a coronal seal. Rubber dam is essential.



Figure 5.24B Patient seen in Figure 5.22 with upper right central incisor having undergone non-vital bleaching and composite addition. The definitive composite should be placed after bleaching for accurate colour matching. Note the dehydration and whitening of adjacent teeth following rubber dam placement.

Whitening strips and topically applied systems

Over-the-counter products are available outside many parts of Europe and over the internet. Their efficacy is based on the in-surgery techniques and these products have proven to be successful. It should be borne in mind, however, that there are always dangers with patient-applied products that are not prescribed and supervised by a dentist.

ACID MICROABRASION

Acid abrasion or microabrasion is essentially a simultaneous process of erosion and abrasion used to remove superficial defects and discolouration within enamel. Strong acids (e.g. 18% hydrochloric acid) and pumice slurry are applied to the enamel and gently agitating using special rubber agitators similar to a rubber point or a rotating rubber cup in a speed-reducing handpiece (to reduce the risk of acid splatter). The acid dissolves the outer layer of enamel which is then worn away by the agitated pumice. The technique is only suitable for superficial defects in enamel such as mild fluorosis; it should not be used for defects in dentine. The process is essentially a controlled way of removing the surface layer of enamel and when 18% hydrochloric acid and pumice are used in this way for approximately 100 seconds, about 360 $\pm$ 130 µm of enamel thickness is lost (Tong et al., 1993). The obvious major issues of using concentrated hydrochloric acid in the mouth are safety should the product be spilt onto the soft tissues; rubber dam is therefore mandatory (Figure 5.25).



Figure 5.25 Patient seen in Figure 5.7A with fluorosis where microabrasion was planned. Rubber dam isolation is essential when carrying out microabrasion. Note the use of floss ligatures to retract the rubber dam, ensuring complete exposure of the crown. Microabrasion failed to improve the appearance in this case.

SUMMARY

This chapter has detailed the most common causes that patients complain about in relation to the appearance of their teeth. Carefully listening to the patient's issues will direct the dentist to the chief complaint. Planning treatment in patients with a heightened concern about their appearance is often difficult and a combined orthodontic restorative approach is often necessary for more complicated cases. This chapter has also dealt with minimally invasive methods of addressing various aspects of appearance such as bleaching for discolouration and microabrasion for superficial enamel defects. More advanced operative methods of dealing with issues of appearance such as the placement of veneers and crowns are dealt with in the relevant subsequent chapters.

FURTHER READING

Brunton, P.A., Aminian, A., Pretty, I.A., 2006. Vital tooth bleaching in dental practice: 2. Novel bleaching systems. Dent. Update 33, 357–358, 360–362.

Pretty, I.A., Ellwood, R.P., Brunton, P.A., Aminian, A., 2006. Vital tooth bleaching in dental practice: 1. Professional bleaching. Dent. Update 33, 288–290, 293–296, 299–300.

Pretty, I.A., Brunton, P., Aminian, A., Davies, R.M., Ellwood, R.P., 2006. Vital tooth bleaching in dental practice: 3. Biological, dental and legal issues. Dent. Update 33, 422–424, 427–428, 431–432.

Tong, L.S., Pang, M.K., Mok, N.Y., King, N.M., Wei, S.H., 1993. The effects of etching, micro-abrasion, and bleaching on surface enamel. J. Dent. Res. 72, 67–71.